

MT 203: Materials Synthesis & Characterization

Course Code	MT 203
Course Name	Materials Synthesis & Characterization
L-T-P-C	3-2-0-4
Prerequisites	None
Intended for	B.Tech. (MSE)
Mutual Exclusion	EN 512 - Structure-Property correlation in materials for Energy Applications.
Approval	57th BoA (Oct 2024)

Course Contents

- **Module-I:** Introduction to the importance of materials synthesis. Describe the different synthesis methods. Describe the process of single crystal growth methods in solution. [4 Hours]
- **Module-II:** Importance of materials characterization. Basic Crystallography; lattice, reciprocal lattice, crystal planes, and directions; Symmetry operations, point and space groups, Structure of Materials – Crystal structure, microstructure, and macrostructure; Determination of crystal structure by diffraction, X-ray diffraction, and electron and neutron diffraction; Vibrational spectroscopy (IR and Raman spectroscopy) for structural characterization of materials. [14 Hours]
- **Module-III:** Microstructure determination by light, and electron microscopy (SEM and TEM); contrast mechanisms in imaging, binary alloys and distribution of phases in microstructure. [6 Hours]
- **Module-IV:** Thermal analysis by TGA-DSC; Determination of enthalpy, melting, decomposition, and phase transition temperatures. [3 Hours]
- **Module-V:** Optical properties of materials, Absorption (UV-Vis), emission (Photoluminescence), Ellipsometry for refractive index. [5 Hours]

Laboratory Modules (teaching laboratory: 24 hours)

1. **Synthesis of materials** (Module-I):
 - (a) Synthesis of materials using hydrothermal, co-precipitation, and sol-gel methods. Types of materials; a. porous ceramic materials; b. composites, c. metal oxides, and d. semiconducting materials
 - (b) 2. Demonstration of single crystal growth in the solution phase.
2. **Structural Characterization** (Module-II):
 - (a) The structural characterization of developed materials using XRD. Determination of crystal structure by X-ray diffraction in a diffractometer, Crystallite size.
 - (b) The determination of functional groups and phase identification of mentioned materials using FT-IR, and Raman spectroscopy.
3. **Microstructural Characterization** (Module-III):
 - (a) The processing of specimen and observation of microstructure under optical microscope, scanning electron microscope.
 - (b) The processing of specimen and observation of microstructure and electron diffraction using a transmission electron microscope.
4. **Optical Characterization** (Module-IV):
 - (a) The processing of specimen for Absorption (UV-Vis), and emission (Photoluminescence/fluorescence) spectroscopy. Thermal Analysis (Module-V):
 - (b) Thermal characterization for the determination of melting, decomposition, phase transition temperatures and enthalpy.

Text Books

1. L.E. Smart and E.A., **Solid State Chemistry: An Introduction**, Taylor & Francis.
2. B. D. Cullity, **Elements of X-Ray Diffraction**.
3. William D. Callister (Jr.), **Materials Science and Engineering**.
4. Anthony West, **Solid State Chemistry and Its Applications**, John Wiley & Sons.

References

1. Marc J.Madou, **Fundamentals of Microfabrication and Nanotechnology**.
2. A. J. Moulson and J. M. Herbert, **Electroceramics: Materials, Properties and Applications**, 2nd edition, Wiley, 2003.
3. V. R. Remya, **Nanostructured Smart Materials Synthesis, Characterization, and Potential Applications**, CRC Press, 2022.
4. M. A. Serio, D. M. Gruen, R. Malhotra, **Synthesis and Characterization of Advanced Materials**, American Chemical Society, 1997.